

Disordered $J - J'$ Heisenberg Antiferromagnet on a Triangular Lattice

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The effect of quenched bond disorder in the triangular lattice Heisenberg antiferromagnet has attracted a lot of attention lately. Recent experiments on crystals which are believed to realize this model have suggested emergence of a glassy phase when the exchange interactions between the spin degrees of freedom are sufficiently disordered. It is also understood that in the presence of bond disorder the 120° non-collinear chiral order is destroyed at $d = 2$ lower critical dimension, although, with a correlation length that scales exponentially with the inverse power of the width of the bond disorder. This rules out the possibility to characterize the system without carefully eliminating the finite size effects. To this end we resort to two different microscopic modelling of the disordered $J_1 - J_2$ Heisenberg hamiltonian, improving on the system size limitations of prior calculations and also mapping out the entire phase diagram of the model which corroborates to the signatures observed in experiments.

I. INTRODUCTION

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II. THE DISORDERED $J - J'$ HEISENBERG HAMILTONIAN ON A TRIANGULAR LATTICE

In a disordered triangular lattice the $J - J'$ antiferromagnetic Heisenberg model has the familiar form,

$$H = \sum_{\langle i,j \rangle} J_{i,j} \vec{S}_i \cdot \vec{S}_j + \sum_{\langle\langle i,j \rangle\rangle} \alpha J'_{i,j} \vec{S}_i \cdot \vec{S}_j \quad (1)$$

with the symbols $\langle i,j \rangle$ and $\langle\langle i,j \rangle\rangle$ representing sum over nearest and next to nearest neighbours. This is a SU(2) symmetric hamiltonian but the discrete translational symmetry of the lattice is lost as the couplings $J_{i,j}$ and $J'_{i,j}$ are all uncorrelated random variables drawn from a random distribution $P(J, \Delta)$ with Δ , the standard deviation characterizing the strength of the disorder. Δ is an implicit parameter for the disordered hamiltonian. In a clean system, however, all the couplings are fixed to a scale and in the absence of any external magnetic field, α becomes the sole adjustable parameter that determines the relative strength of the next to nearest neighbour (NNN) couplings over the nearest neighbour (NN) couplings. In the following we shall consider both α and Δ as parameters of the model and explore the consequent phase diagram.

For $\alpha = 0$ the ground state of the homogeneous triangular lattice Heisenberg antiferromagnet has a non-collinear order characterized by the 120° spiral spin orientation with an ordering wave vector $\vec{Q} = (2\pi/3, 0)$. This is a well known and well studied example of the classical Néel order surviving the effects of quantum fluctuations¹. By tuning up α

one can see that the classical ground state of this model goes through a phase transition from the three sublattice 120° state to a degenerate continuum of four sublattice stripy states at $\alpha = 1/8$ ^{2,3}. Quantum effects become crucial near the phase transition point and it is believed that spin liquid like states emerge interposed between the non-collinear 120° and the **collinear** stripe phase^{3,4}.

The introduction of bond disorder makes this model particularly interesting as the non-uniformity in the coupling profile tries to compete with and unwind the effects of the underlying geometric frustration of the lattice and thereby makes further diversification of the phase map plausible. It is known that in a non-collinear antiferromagnet, long range Néel order is destroyed by random inhomogeneity in the exchange interactions below or at the critical dimension $d = 2$ ⁵. Our planar $J - J'$ heisenberg antiferromagnet on a triangular lattice for $\alpha < 1/8$ is precisely an example of such a system placed at the critical dimension. With the tentative absence of long range antiferromagnetic order in the presence of quenched bond disorder^{5,6}, the weakly frustrated triangular lattice antiferromagnet is amenable to a spin-glass transition⁷. In the analysis that follows, we show that quantum fluctuations disorders the system even further to restrict the surviving short range correlations of the neighbouring spins on the lattice. The fact that quantum effects also become very important in tandem with disorder makes way for a possible departure from the classical degrees of freedom when the effects of quantum and quenched disorder is particularly strong. Our scheme, split between two complimentary approaches, one considering spin-wave fluctuations on top of the classical ground state of the disordered lattice and another being a slave-fermion mean field approximation capturing excitations which are inherently quantum in nature, explores the microscopic description of the ground state of this model from the onset of the breakdown of the long range antiferromagnetic order to the limits of very strong disorder.

III. SPINWAVE EXCITATIONS WITH FINITE DISORDER

The spin-wave theory (SWT) for a disordered Heisenberg hamiltonian, not unlike its homogeneous counterpart, entails

an expansion in terms of quantum excitation modes around the magnetically ordered classical ground state^{8,9}. The starting point for such an expansion is the representation of the SU(2) symmetric spin operators at each lattice site with slave bosons (e.g. Holstein-Primakoff or Schwinger bosons) where the quantization axes are chosen to align with the orientation of each individual spin, as dictated by the classical lowest energy configuration¹⁰. Although the transformations themselves are exact, they retain the highly interacting characteristic of the heisenberg model. The spin-wave approximation, i.e. the assumption that the quantum corrections due to these bosons are weaker compared to the corresponding classical expectation values is an additional input which allows a controlled expansion of the hamiltonian in terms of these bosonic modes. The chief distinction in the disordered case is that the loss of the lattice translational symmetry prevents any description in terms of lattice momentums, which are no longer good quantum numbers and instead a real space formulation is necessitated.

The situation is further complicated by the non-triviality of the energy landscape of the disordered classical hamiltonian. In our treatment we follow a simple local energy minimization scheme iteratively to reach the ground state configuration. The scheme involves a starting with a randomized initial spin configurations and afterwards performing local energy minimization at each site to antialign the spins with respect to the net local field coupled with it until a threshold of accuracy is achieved^{11,12}. Although this method is inherently local in nature, after a sufficient amount of sampling dispenses with the outliers the typical solutions do get very close to the true ground states in terms of the relevant small wavelength features^{12,13}.

The classical ground state obtained through the iterative algorithm may be interpreted as a configuration of rotation matrices R_i describing the passive rotations of individual spins with respect to the laboratory frame, $\vec{S}_i = R_i \cdot \vec{S}_i^l$. This picture is useful for performing the spin-wave expansion around the quantization axis S_i^z attached to the body frame of the spins. The rotation matrices themselves can be expressed with pairs of Euler angles θ_i, ϕ_i describing the rigid body rotation of the body axis.

$$R_i = \begin{pmatrix} \cos \theta_i \cos \phi_i & -\sin \phi_i & \sin \theta_i \cos \phi_i \\ \cos \theta_i \sin \phi_i & \cos \phi_i & \sin \theta_i \sin \phi_i \\ -\sin \theta_i & 0 & \cos \theta_i \end{pmatrix} \quad (2)$$

For notational simplicity the hamiltonian in Eq. (1) can be recast in the form $H = 1/2 \sum_{i,j} J_{i,j} \vec{S}_i \cdot \vec{S}_j$, where both the parameters α and Δ are implicit and they control the magnitudes of the couplings $J_{i,j}$ which are otherwise zero unless $(i,j) \in \text{NN or NNN lattice points}$. In the rotated frame the hamiltonian reads,

$$H = \frac{1}{2} \sum_{i,j} J_{i,j} (R_i^{-1} \cdot R_j)_{\alpha\beta} S_i'^\alpha S_j'^\beta \quad (3)$$

with the standard Holstein-Primakoff (HP) bosons introduced in a very assymmetric way,

$$\begin{aligned} S_i'^z &= S - a_i^\dagger a_i \\ S_i'^+ &= \sqrt{2S - a_i^\dagger a_i} a_i \\ S_i'^- &= a_i^\dagger \sqrt{2S - a_i^\dagger a_i} \end{aligned} \quad (4)$$

In a disordered lattice of a finite size N , one has to consider all these N bosonic operators, a_i s, together while performing the spin-wave approximation of the hamiltonian (3) in real space. The consideration of a $2N$ dimensional vector $\vec{\gamma} = (\vec{a}, \vec{a}^\dagger)$ consisting of the HP bosons and their hermitian conjugates offers a relative simplification of the resulting expression. The expansion, as obtained for the Heisenberg model on a generic lattice upto third order¹⁴, can be neatly collected in the powers of the γ_i operators.

$$\begin{aligned} H &= H^0 + \sum_i^{2N} K_i^{(1)} \gamma_i + \sum_{i,j}^{2N} K_{i,j}^{(2)} \gamma_i \gamma_j \\ &+ \sum_{i,j,k}^{2N} K_{i,j,k}^{(3)} \gamma_i \gamma_j \gamma_k + \mathcal{O}(\gamma^4) \end{aligned} \quad (5)$$

with the coefficients given by,

$$\begin{aligned} H_0 &= \frac{S^2}{2} \sum_{i,j}^N J_{i,j} [\cos \theta_i \cos \theta_j + \cos(\phi_i - \phi_j) \sin \theta_i \sin \theta_j] \\ K_i^{(1)} &= \begin{cases} \Gamma_i^{(1)} = \frac{S^{3/2}}{\sqrt{2}} \sum_j^N J_{i,j} [\cos(\phi_i - \phi_j) \cos \theta_i \sin \theta_j - \sin \theta_i \cos \theta_j - i \sin(\phi_i - \phi_j) \sin \theta_j], & \text{if } i \leq N \\ \Gamma_{i-N}^{(1)*}, & \text{if } i > N \end{cases} \\ K_i^{(2)} &= \begin{cases} \frac{S^{3/2}}{\sqrt{2}} \sum_j^N J_{i,j} [\cos(\phi_i - \phi_j) \cos \theta_i \sin \theta_j - \sin \theta_i \cos \theta_j - i \sin(\phi_i - \phi_j) \sin \theta_j], & \text{if } i \leq N \\ \frac{S^{3/2}}{\sqrt{2}} \sum_j^N J_{i,j} [\cos(\phi_i - \phi_j) \cos \theta_i \sin \theta_j - \sin \theta_i \cos \theta_j - i \sin(\phi_i - \phi_j) \sin \theta_j], & \text{if } i > N \end{cases} \end{aligned} \quad (6)$$

To do: finish equation

When the expansion is around the true ground state of a **homogeneous** lattice the linear term in the hamiltonian (7) drops out to leading order¹⁵. This relates to the absence of transverse or angular fluctuation of the classically ordered ground state. The spin-wave excitations, however, renormalizes the staggered moment heavily¹ and in principle can further enhance the effect of quenched disorder. In addition, for an expansion around the glassy ground state of the disordered spin model one can not, apriori, rule out the perturbative corrections to classical angles. In our treatment we follow a strictly $1/S$ expansion scheme and consider both the third order term and the linear term of the hamiltonian (7) together and find the appropriate angle correction to the classical ground state at finite disorder in a saddle point approximation. With the quantum corrected angles, a generalized bogolioubov transformation **To do: Bogolioubov transformation** brings the quadratic part of the hamiltonian in a diagonal form⁸ while the linear and the cubic terms now identically vanishes at the mean field level.

To do: Bogolioubov transformation equations

The angle corrections are obtained from solving the following equations.

To do: angle correction equation

It is natural to expect that for relatively weaker disorder i.e. when the ground state is close to the 120° spiral state, the success of the linear spin wave theory (i.e. effectively truncating the spin-wave expansion after the second order) in describing the appropriate quantum ground state for the model would be repeated. However, the outcome of such a calculation is limited to

A brief discussion of why we might want to go beyond the spin-waves

IV. MEAN-FIELD THEORY IN REAL SPACE

Singlet formation is usually favored in highly disordered antiferromagnetic systems^{16,17}. However, in the complicated

energy landscape of our model which is both frustrated and disordered, a competition between spin glass type and valence bond type behavior can be expected. To that effect we consider a composite mean-field expansion of our model hamiltonian (1) which consists of the slave fermionic Hubbard-Stratanovich transformation in the spirit of a Large-N expansion¹⁷ but then goes onto include the magnetic moments of an Weiss mean-field theory. The mean-field hamiltonian assumes the following form,

$$H_{\text{mf}} = - \sum_{\langle i,j \rangle, \alpha} \left(\chi_{ij}^* f_{i\alpha}^\dagger f_{j,\alpha} + \text{h.c.} \right) + \frac{1}{2} \sum_{i, \alpha, \beta} \vec{B}_i \cdot f_{i\alpha}^\dagger \vec{\sigma}_{\alpha, \beta} f_{i\beta} \quad (7)$$

The mean-field parameters are determined by the following equations.

mean-field equations

Although both these channels can not be accessed together by a single canonical transformation the inclusion of both the magnetic and bond instabilities in a single mean-field theory is useful demonstrates the competition between the two relevant phases of our disordered model.

In the absence of the bond parameters our mean-field theory is equivalent to a problem of minimization of energy for classical spins in a disordered media. In essence this is the same starting point we use to find the spin-wave excitation for the semi-classical analysis. For the mean-field hamiltonian, quantum suppression of the spin moments can lead to a variety of singlet ground state depending upon the implicit parameters of our model (1), α and Δ .

V. RESULTS

VI. CONCLUSIONS

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